

In [1]: `import pandas as pd`

```
df1 = pd.read_csv('VAE_ranking_evaluation_metrics_1.csv')
df2 = pd.read_csv('VAE_ranking_evaluation_metrics_2.csv')

merged_df = pd.concat([df1, df2], ignore_index=True)

merged_df.to_csv('VAE_ranking_evaluation_metrics_merged.csv', index=False)
```

In [7]: `import pandas as pd`
`import matplotlib.pyplot as plt`
`import seaborn as sns`

```
# Load data
gan_df = pd.read_csv("GAN_ranking_evaluation_metrics.csv")
vae_df = pd.read_csv("VAE_ranking_evaluation_metrics.csv")

gan_df['Model'] = 'GAN'
vae_df['Model'] = 'VAE'

combined_df = pd.concat([gan_df, vae_df], ignore_index=True)
melted_df = combined_df.melt(id_vars=['Model'], var_name='Metric', value_name='Score')

melted_df['Metric'] = melted_df['Metric'].astype(str)
melted_df['Score'] = pd.to_numeric(melted_df['Score'], errors='coerce')
melted_df = melted_df.dropna(subset=['Score'])

plt.figure(figsize=(14, 7))
sns.set_style("whitegrid")

# Use a softer pastel palette
palette = sns.color_palette("pastel")

# Barplot with dodge for side-by-side bars
barplot = sns.barplot(
    data=melted_df,
    x='Metric',
    y='Score',
    hue='Model',
    palette=palette,
    ci='sd',
    capsize=0.1,
    edgecolor='gray'
)

# Overlay individual data points for each model per metric
sns.stripplot(
    data=melted_df,
    x='Metric',
    y='Score',
    hue='Model',
    dodge=True,
    marker='o',
    alpha=0.6,
```

```

    palette=palette,
    edgecolor='gray',
    linewidth=0.5,
    size=6
)

# Remove duplicate legends (keep only one)
handles, labels = barplot.get_legend_handles_labels()
plt.legend(handles[0:2], labels[0:2], title='Model', loc='upper right')

plt.title('Comparison of Semantic Consistency Evaluation Metrics\nGAN vs VAE')
plt.xlabel('Metric', fontsize=14)
plt.ylabel('Score', fontsize=14)
plt.xticks(rotation=45, ha='right', fontsize=12)
plt.yticks(fontsize=12)
plt.tight_layout()
plt.show()

```

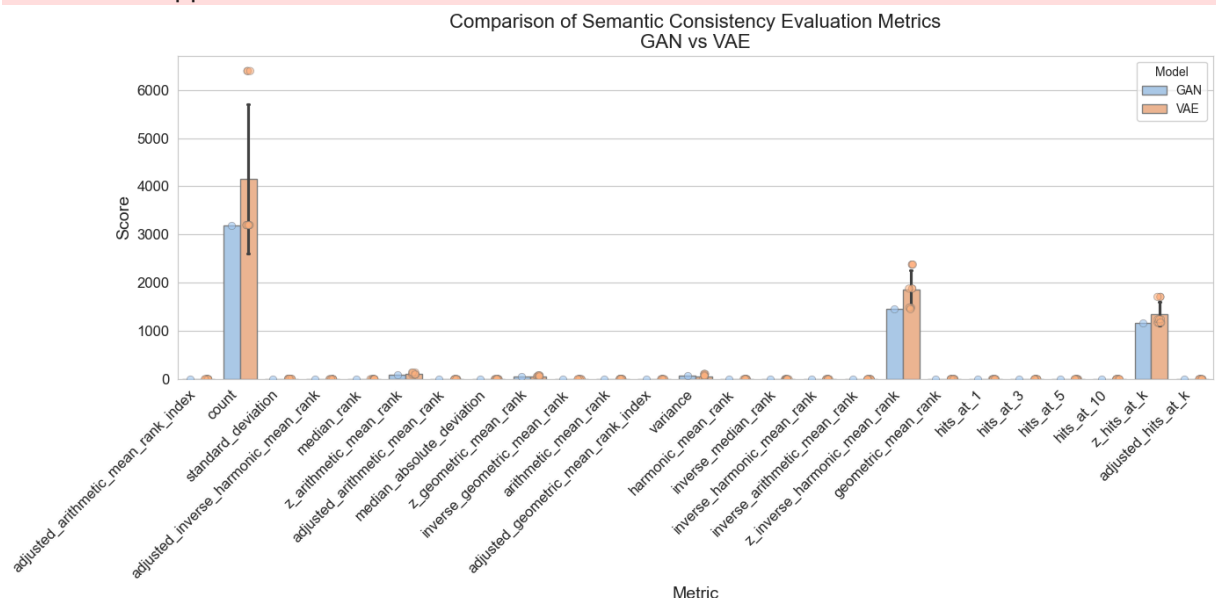
/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_74237/1725503171.py:26: FutureWarning:

The `ci` parameter is deprecated. Use `errorbar='sd'` for the same effect.

```

barplot = sns.barplot(
/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_74237/1725503171.py:26: UserWarning: The palette list has more values (10) than needed (2), which may not be intended.
barplot = sns.barplot(
/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_74237/1725503171.py:38: UserWarning: The palette list has more values (10) than needed (2), which may not be intended.
sns.stripplot(
/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_74237/1725503171.py:38: FutureWarning: Use "auto" to set automatic grayscale colors. From v0.14.0, "gray" will default to matplotlib's definition.
sns.stripplot(

```



```

In [10]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load data
gan_df = pd.read_csv("GAN_ranking_evaluation_metrics.csv")
vae_df = pd.read_csv("VAE_ranking_evaluation_metrics.csv")

gan_df['Model'] = 'GAN'
vae_df['Model'] = 'VAE'

combined_df = pd.concat([gan_df, vae_df], ignore_index=True)

# Convert to long format
melted_df = combined_df.melt(id_vars=['Model'], var_name='Metric', value_name='Score')
melted_df['Metric'] = melted_df['Metric'].astype(str)
melted_df['Score'] = pd.to_numeric(melted_df['Score'], errors='coerce')
melted_df = melted_df.dropna(subset=['Score'])

# Define metric groups:
group1 = ['count', 'z_inverse_harmonic_mean_rank', 'z_hits_at_k']
group2 = ['variance', 'z_arithmetic_mean_rank', 'z_geometric_mean_rank']
group3_start = ['adjusted_arithmetic_mean_rank', 'hits_at_1']
# Add all starting with 'inverse_' to group 3
inverse_metrics = [m for m in melted_df['Metric'].unique() if m.startswith('inverse_')]
group3 = group3_start + inverse_metrics

# Combine groups to find remaining metrics
all_metrics = set(melted_df['Metric'].unique())
used_metrics = set(group1 + group2 + group3)
remaining_metrics = list(all_metrics - used_metrics)
remaining_metrics.sort()

# Split remaining metrics into two halves
half = len(remaining_metrics) // 2
remaining_group1 = remaining_metrics[:half]
remaining_group2 = remaining_metrics[half:]

# Function to plot one group of metrics
def plot_metric_group(metric_list, title):
    plot_df = melted_df[melted_df['Metric'].isin(metric_list)]
    plt.figure(figsize=(12, 6))
    sns.set_style("whitegrid")
    palette = sns.color_palette("pastel")

    barplot = sns.barplot(
        data=plot_df,
        x='Metric',
        y='Score',
        hue='Model',
        palette=palette,
        ci='sd',
        capsize=0.1,
        edgecolor='gray'
    )

```

```

sns.stripplot(
    data=plot_df,
    x='Metric',
    y='Score',
    hue='Model',
    dodge=True,
    marker='o',
    alpha=0.6,
    palette=palette,
    edgecolor='gray',
    linewidth=0.5,
    size=6
)
# Remove duplicate legends
handles, labels = barplot.get_legend_handles_labels()
plt.legend(handles[0:2], labels[0:2], title='Model', loc='upper right')

plt.title(title, fontsize=16)
plt.xlabel('Metric', fontsize=14)
plt.ylabel('Score', fontsize=14)
plt.xticks(rotation=45, ha='right', fontsize=12)
plt.yticks(fontsize=12)
plt.tight_layout()
plt.show()

# Plot each group
plot_metric_group(group1, "Group 1: count, z_inverse_harmonic_mean_rank, z_h")
plot_metric_group(group2, "Group 2: variance, z_arithmetic_mean_rank, z_geom")
plot_metric_group(group3, "Group 3: adjusted_arithmetic_mean_rank, hits_at_1")
plot_metric_group(remaining_group1, "Remaining Metrics - Part 1")
plot_metric_group(remaining_group2, "Remaining Metrics - Part 2")

```

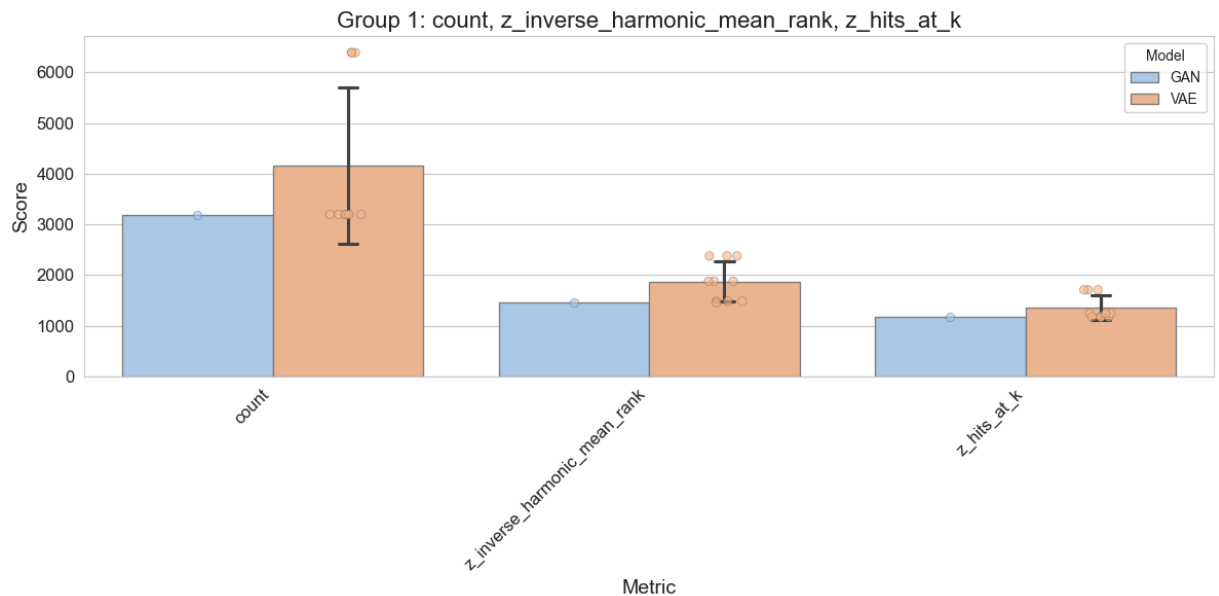
/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_74237/2942174955.
py:46: FutureWarning:

The `ci` parameter is deprecated. Use `errorbar='sd'` for the same effect.

```

barplot = sns.barplot(
/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_74237/2942174955.
py:46: UserWarning: The palette list has more values (10) than needed (2), w
hich may not be intended.
    barplot = sns.barplot(
/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_74237/2942174955.
py:56: UserWarning: The palette list has more values (10) than needed (2), w
hich may not be intended.
    sns.stripplot(
/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_74237/2942174955.
py:56: FutureWarning: Use "auto" to set automatic grayscale colors. From v0.
14.0, "gray" will default to matplotlib's definition.
    sns.stripplot(

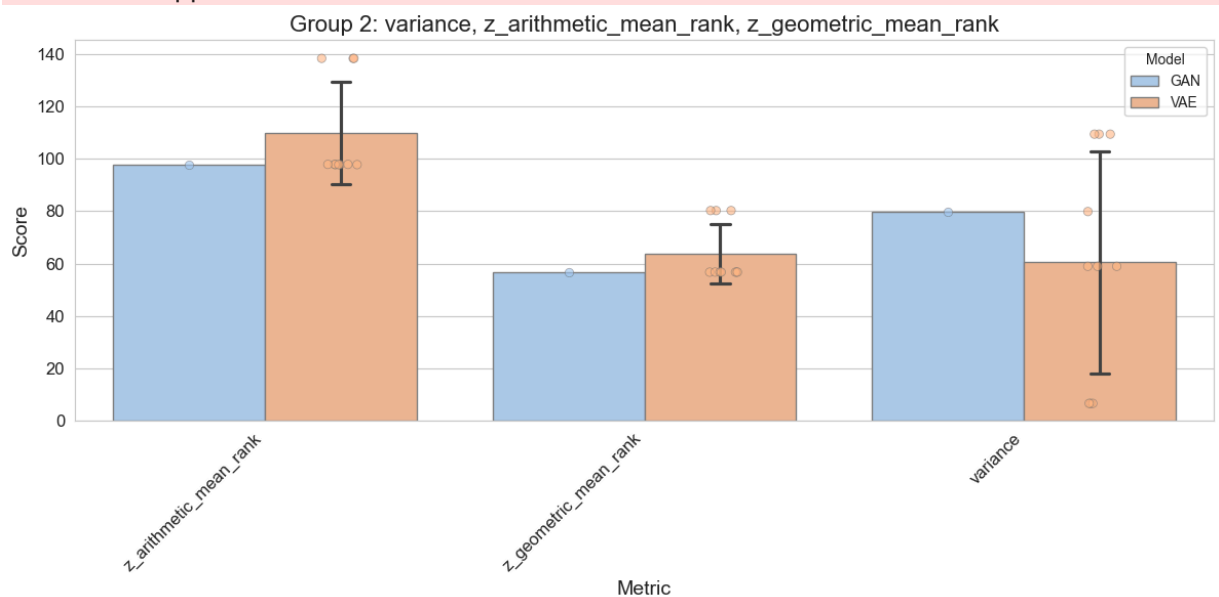
```



```
/var/folders/by/r1y5wccx25jfdv7nvpm9pf40000gn/T/ipykernel_74237/2942174955.py:46: FutureWarning:
```

The `ci` parameter is deprecated. Use `errorbar='sd'` for the same effect.

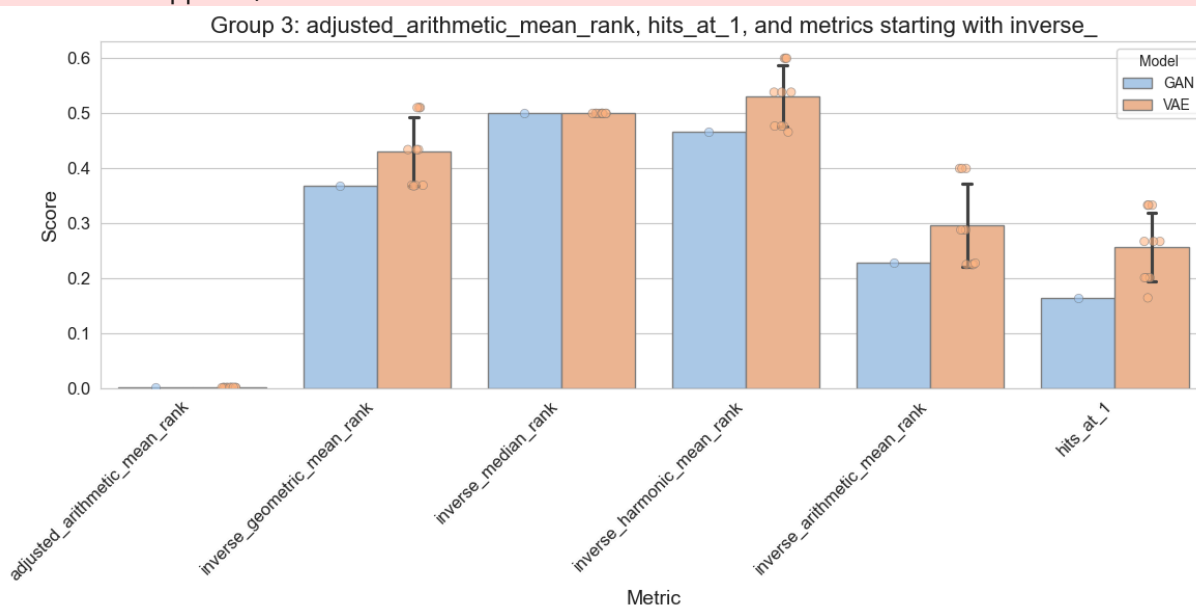
```
barplot = sns.barplot(
/var/folders/by/r1y5wccx25jfdv7nvpm9pf40000gn/T/ipykernel_74237/2942174955.py:46: UserWarning: The palette list has more values (10) than needed (2), which may not be intended.
barplot = sns.barplot(
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sns.stripplot(
/var/folders/by/r1y5wccx25jfdv7nvpm9pf40000gn/T/ipykernel_74237/2942174955.py:56: FutureWarning: Use "auto" to set automatic grayscale colors. From v0.14.0, "gray" will default to matplotlib's definition.
sns.stripplot(
```



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/var/folders/by/r1y5wccx25jfdv7nvpm9pf40000gn/T/ipykernel_74237/2942174955.py:46: FutureWarning:
```

The `ci` parameter is deprecated. Use `errorbar='sd'` for the same effect.

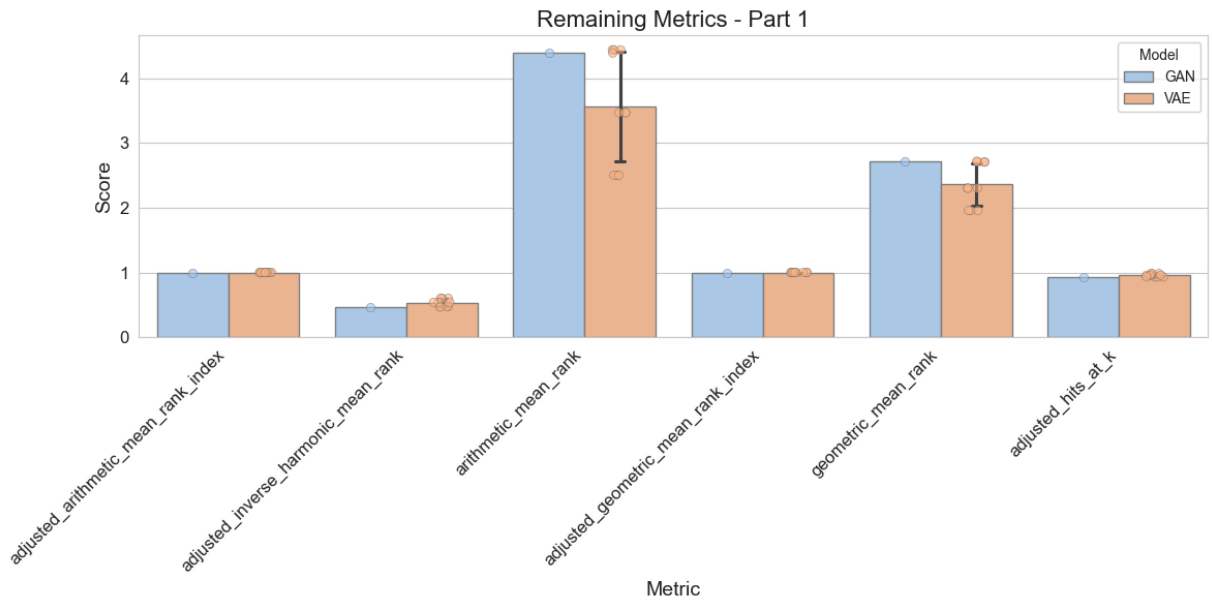
```
barplot = sns.barplot(
/var/folders/by/r1y5wccx25jfdv7nvpm9pf40000gn/T/ipykernel_74237/2942174955.py:46: UserWarning: The palette list has more values (10) than needed (2), which may not be intended.
barplot = sns.barplot(
/var/folders/by/r1y5wccx25jfdv7nvpm9pf40000gn/T/ipykernel_74237/2942174955.py:56: UserWarning: The palette list has more values (10) than needed (2), which may not be intended.
sns.stripplot(
/var/folders/by/r1y5wccx25jfdv7nvpm9pf40000gn/T/ipykernel_74237/2942174955.py:56: FutureWarning: Use "auto" to set automatic grayscale colors. From v0.14.0, "gray" will default to matplotlib's definition.
sns.stripplot(
```



```
/var/folders/by/r1y5wccx25jfdv7nvpm9pf40000gn/T/ipykernel_74237/2942174955.py:46: FutureWarning:
```

The `ci` parameter is deprecated. Use `errorbar='sd'` for the same effect.

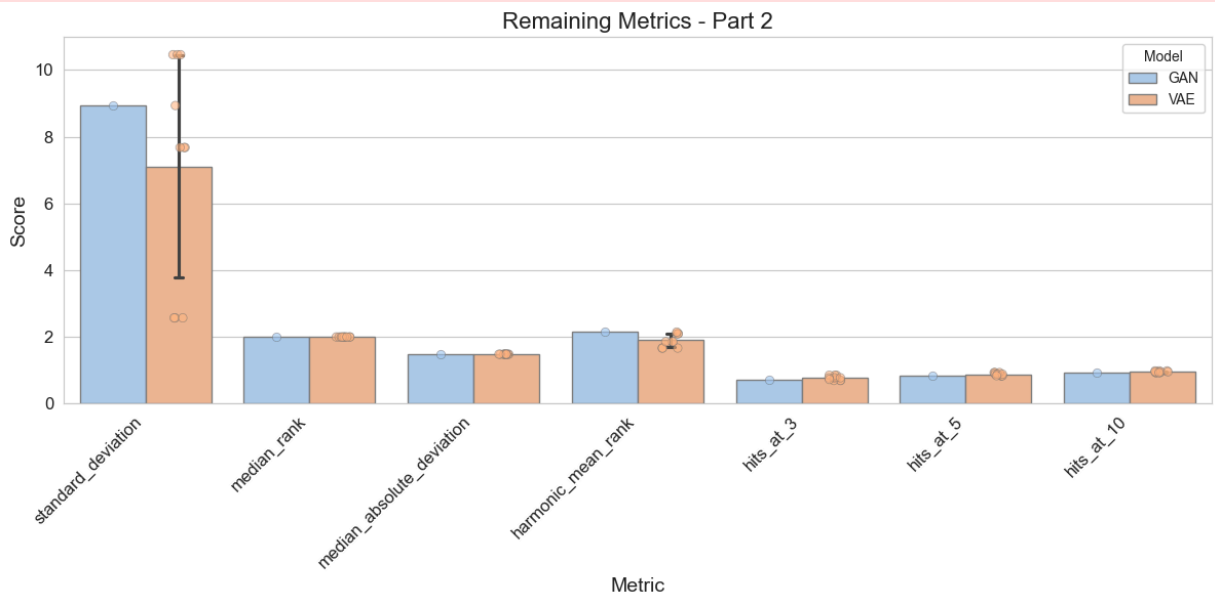
```
barplot = sns.barplot(
/var/folders/by/r1y5wccx25jfdv7nvpm9pf40000gn/T/ipykernel_74237/2942174955.py:46: UserWarning: The palette list has more values (10) than needed (2), which may not be intended.
barplot = sns.barplot(
/var/folders/by/r1y5wccx25jfdv7nvpm9pf40000gn/T/ipykernel_74237/2942174955.py:56: UserWarning: The palette list has more values (10) than needed (2), which may not be intended.
sns.stripplot(
/var/folders/by/r1y5wccx25jfdv7nvpm9pf40000gn/T/ipykernel_74237/2942174955.py:56: FutureWarning: Use "auto" to set automatic grayscale colors. From v0.14.0, "gray" will default to matplotlib's definition.
sns.stripplot(
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/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_74237/2942174955.py:46: FutureWarning:
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The `ci` parameter is deprecated. Use `errorbar='sd'` for the same effect.

```
barplot = sns.barplot(
/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_74237/2942174955.py:46: UserWarning: The palette list has more values (10) than needed (2), which may not be intended.
barplot = sns.barplot(
/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_74237/2942174955.py:56: UserWarning: The palette list has more values (10) than needed (2), which may not be intended.
sns.stripplot(
/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_74237/2942174955.py:56: FutureWarning: Use "auto" to set automatic grayscale colors. From v0.14.0, "gray" will default to matplotlib's definition.
sns.stripplot(
```



```

In [12]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import os

# Load data
gan_df = pd.read_csv("GAN_ranking_evaluation_metrics.csv")
vae_df = pd.read_csv("VAE_ranking_evaluation_metrics.csv")

gan_df['Model'] = 'GAN'
vae_df['Model'] = 'VAE'

combined_df = pd.concat([gan_df, vae_df], ignore_index=True)

# Convert to long format
melted_df = combined_df.melt(id_vars=['Model'], var_name='Metric', value_name='Score')
melted_df['Metric'] = melted_df['Metric'].astype(str)
melted_df['Score'] = pd.to_numeric(melted_df['Score'], errors='coerce')
melted_df = melted_df.dropna(subset=['Score'])

# Define metric groups:
group1 = ['count', 'z_inverse_harmonic_mean_rank', 'z_hits_at_k']
group2 = ['variance', 'z_arithmetic_mean_rank', 'z_geometric_mean_rank']
group3_start = ['adjusted_arithmetic_mean_rank', 'hits_at_1']
inverse_metrics = [m for m in melted_df['Metric'].unique() if m.startswith('z_')]
group3 = group3_start + inverse_metrics

all_metrics = set(melted_df['Metric'].unique())
used_metrics = set(group1 + group2 + group3)
remaining_metrics = list(all_metrics - used_metrics)
remaining_metrics.sort()

half = len(remaining_metrics) // 2
remaining_group1 = remaining_metrics[:half]
remaining_group2 = remaining_metrics[half:]

# Create directory to save plots
output_dir = "evaluation_plots"
os.makedirs(output_dir, exist_ok=True)

def plot_metric_group(metric_list, title, filename):
    plot_df = melted_df[melted_df['Metric'].isin(metric_list)]
    plt.figure(figsize=(12, 6))
    sns.set_style("whitegrid")
    palette = sns.color_palette("pastel")

    barplot = sns.barplot(
        data=plot_df,
        x='Metric',
        y='Score',
        hue='Model',
        palette=palette,
        ci='sd',
        capsize=0.1,
        edgecolor='gray'
    )

```

```

)
sns.stripplot(
    data=plot_df,
    x='Metric',
    y='Score',
    hue='Model',
    dodge=True,
    marker='o',
    alpha=0.6,
    palette=palette,
    edgecolor='gray',
    linewidth=0.5,
    size=6
)
handles, labels = barplot.get_legend_handles_labels()
plt.legend(handles[0:2], labels[0:2], title='Model', loc='upper right')

plt.title(title, fontsize=16)
plt.xlabel('Metric', fontsize=14)
plt.ylabel('Score', fontsize=14)
plt.xticks(rotation=45, ha='right', fontsize=12)
plt.yticks(fontsize=12)
plt.tight_layout()

# Save figure
plt.savefig(os.path.join(output_dir, filename), dpi=300)
plt.show()

# Plot each group and save images
plot_metric_group(group1, "Group 1: count, z_inverse_harmonic_mean_rank, z_h
plot_metric_group(group2, "Group 2: variance, z_arithmetic_mean_rank, z_geom
plot_metric_group(group3, "Group 3: adjusted_arithmetic_mean_rank, hits_at_1
plot_metric_group(remaining_group1, "Remaining Metrics - Part 1", "remaining
plot_metric_group(remaining_group2, "Remaining Metrics - Part 2", "remaining

print(f"Plots saved to folder: {output_dir}")

```

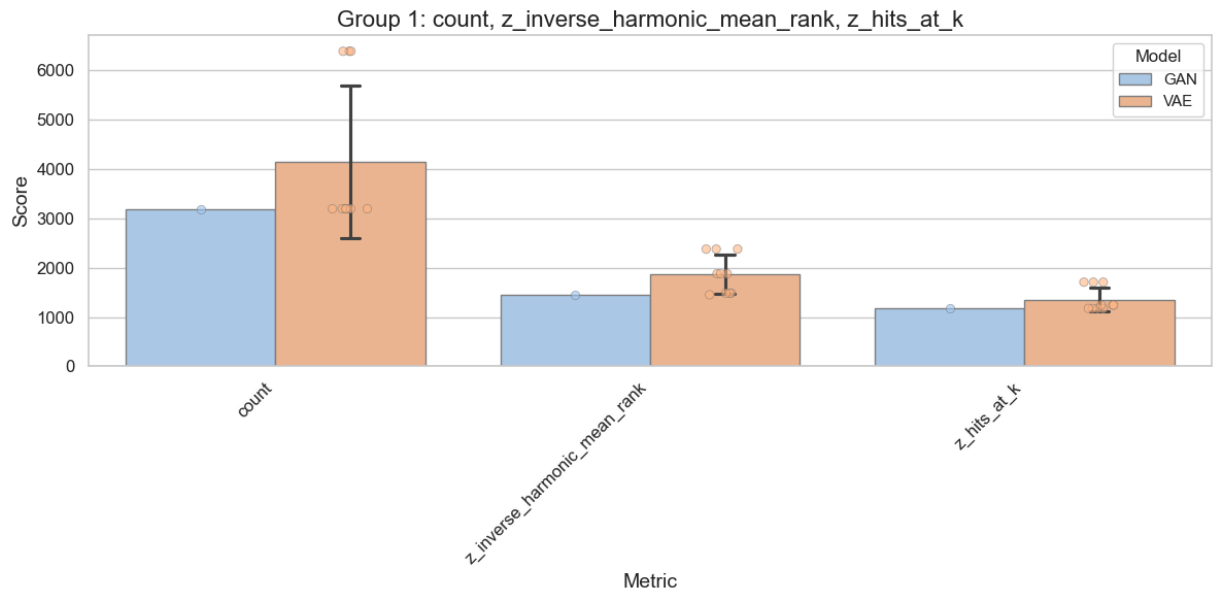
/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_74237/3806083251.
py:47: FutureWarning:

The `ci` parameter is deprecated. Use `errorbar='sd'` for the same effect.

```

barplot = sns.barplot(
/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_74237/3806083251.
py:47: UserWarning: The palette list has more values (10) than needed (2), w
hich may not be intended.
barplot = sns.barplot(
/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_74237/3806083251.
py:57: UserWarning: The palette list has more values (10) than needed (2), w
hich may not be intended.
sns.stripplot(
/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_74237/3806083251.
py:57: FutureWarning: Use "auto" to set automatic grayscale colors. From v0.
14.0, "gray" will default to matplotlib's definition.
sns.stripplot(

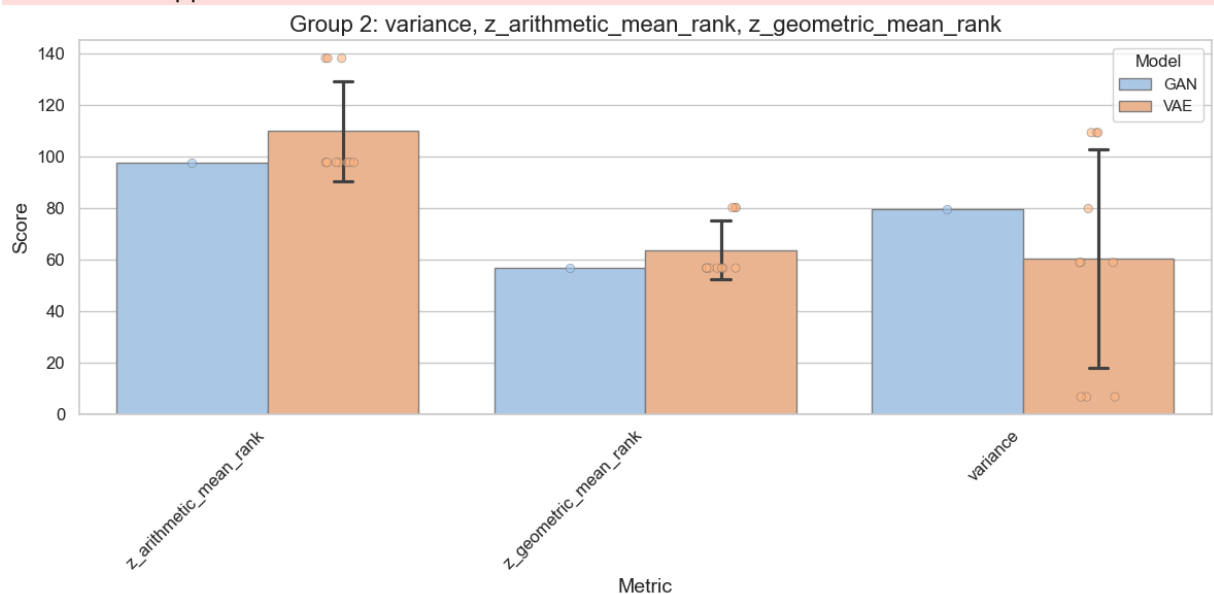
```



```
/var/folders/by/r1y5wccx25jfdv7nvpm9pf40000gn/T/ipykernel_74237/3806083251.py:47: FutureWarning:
```

The `ci` parameter is deprecated. Use `errorbar='sd'` for the same effect.

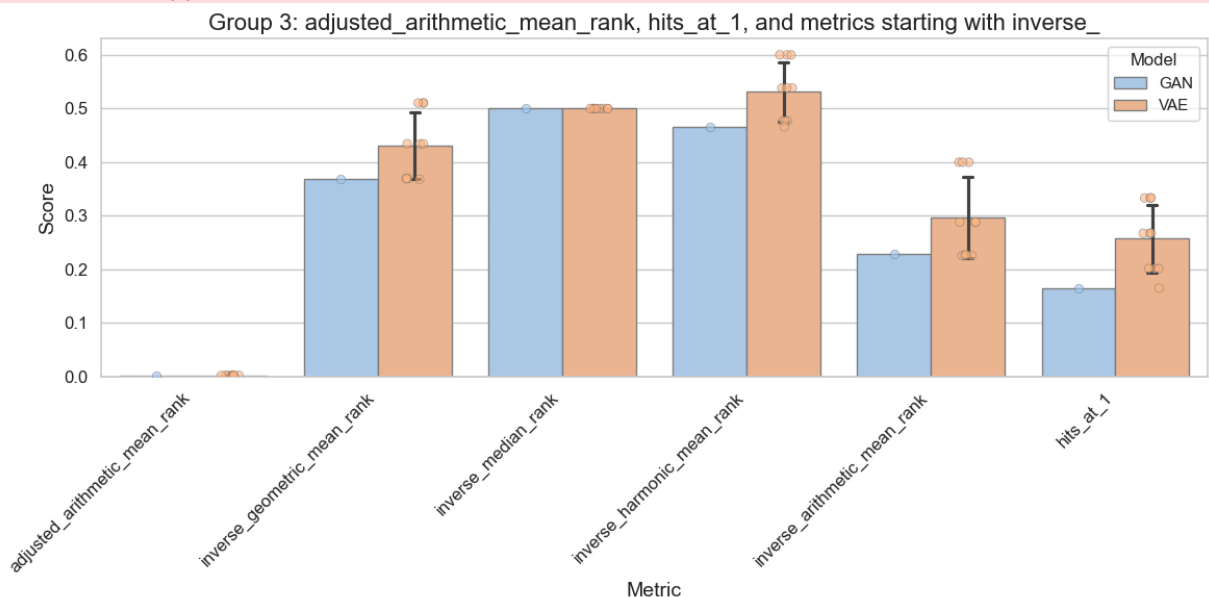
```
barplot = sns.barplot(
/var/folders/by/r1y5wccx25jfdv7nvpm9pf40000gn/T/ipykernel_74237/3806083251.py:47: UserWarning: The palette list has more values (10) than needed (2), which may not be intended.
barplot = sns.barplot(
/var/folders/by/r1y5wccx25jfdv7nvpm9pf40000gn/T/ipykernel_74237/3806083251.py:57: UserWarning: The palette list has more values (10) than needed (2), which may not be intended.
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/var/folders/by/r1y5wccx25jfdv7nvpm9pf40000gn/T/ipykernel_74237/3806083251.py:57: FutureWarning: Use "auto" to set automatic grayscale colors. From v0.14.0, "gray" will default to matplotlib's definition.
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```



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/var/folders/by/r1y5wccx25jfdv7nvpm9pf40000gn/T/ipykernel_74237/3806083251.py:47: FutureWarning:
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The `ci` parameter is deprecated. Use `errorbar='sd'` for the same effect.

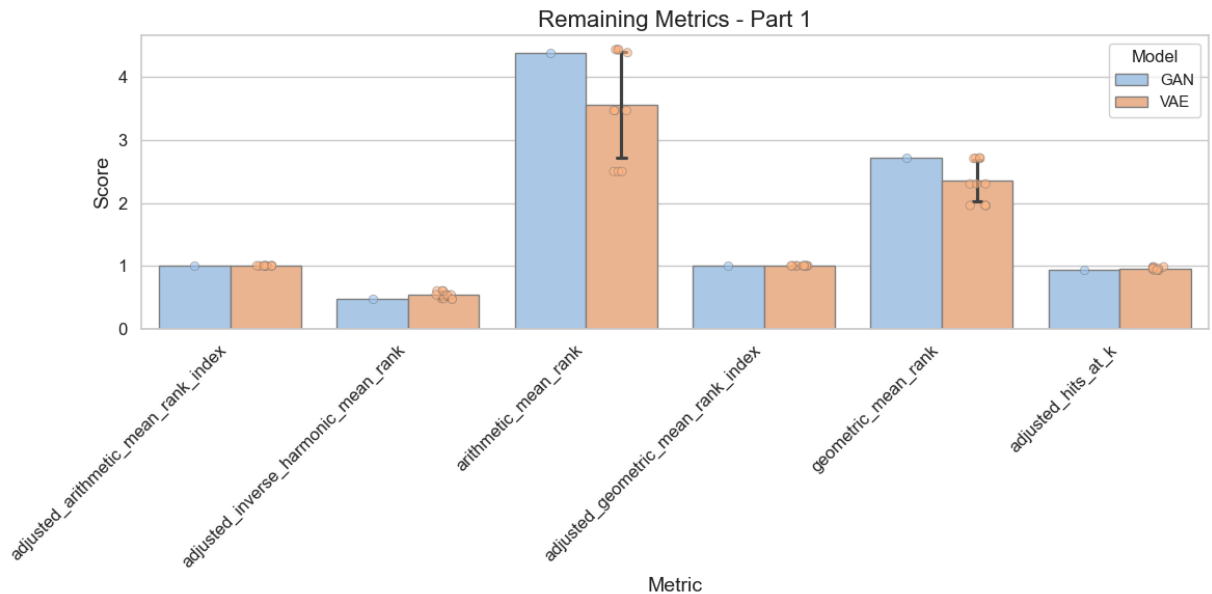
```
barplot = sns.barplot(
/var/folders/by/r1y5wccx25jfdv7nvpm9pf40000gn/T/ipykernel_74237/3806083251.py:47: UserWarning: The palette list has more values (10) than needed (2), which may not be intended.
barplot = sns.barplot(
/var/folders/by/r1y5wccx25jfdv7nvpm9pf40000gn/T/ipykernel_74237/3806083251.py:57: UserWarning: The palette list has more values (10) than needed (2), which may not be intended.
sns.stripplot(
/var/folders/by/r1y5wccx25jfdv7nvpm9pf40000gn/T/ipykernel_74237/3806083251.py:57: FutureWarning: Use "auto" to set automatic grayscale colors. From v0.14.0, "gray" will default to matplotlib's definition.
sns.stripplot(
```



```
/var/folders/by/r1y5wccx25jfdv7nvpm9pf40000gn/T/ipykernel_74237/3806083251.py:47: FutureWarning:
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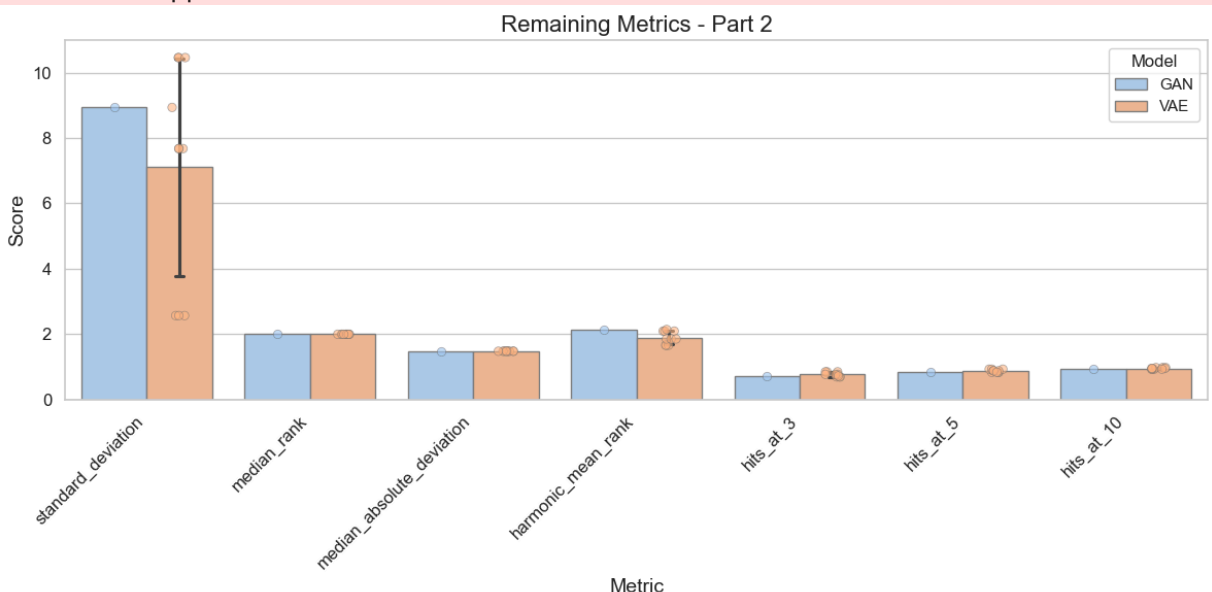
```
barplot = sns.barplot(
/var/folders/by/r1y5wccx25jfdv7nvpm9pf40000gn/T/ipykernel_74237/3806083251.py:47: UserWarning: The palette list has more values (10) than needed (2), which may not be intended.
barplot = sns.barplot(
/var/folders/by/r1y5wccx25jfdv7nvpm9pf40000gn/T/ipykernel_74237/3806083251.py:57: UserWarning: The palette list has more values (10) than needed (2), which may not be intended.
sns.stripplot(
/var/folders/by/r1y5wccx25jfdv7nvpm9pf40000gn/T/ipykernel_74237/3806083251.py:57: FutureWarning: Use "auto" to set automatic grayscale colors. From v0.14.0, "gray" will default to matplotlib's definition.
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```



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/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_74237/3806083251.py:47: UserWarning: The palette list has more values (10) than needed (2), which may not be intended.
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/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_74237/3806083251.py:57: UserWarning: The palette list has more values (10) than needed (2), which may not be intended.
sns.stripplot(
/var/folders/by/r1y5wccx25jfdv7nvpnb9pf40000gn/T/ipykernel_74237/3806083251.py:57: FutureWarning: Use "auto" to set automatic grayscale colors. From v0.14.0, "gray" will default to matplotlib's definition.
sns.stripplot(
```



Plots saved to folder: evaluation_plots